
PhD Candidate in Statistics with strong practical experience in **data preparation, exploratory data analysis, statistical modeling, simulation studies, and report writing**. Proficient in **R, SAS, SQL, and Python**; experienced in collaborating with domain scientists to translate analytical results into actionable recommendations and in applying advanced methods to solve real-world problems, including **insurance pricing, predictive modeling, economic analysis, and biomedical research**.

Technical Proficiencies

Programming: R (mgcv, ggplot2, Shiny), SAS, Python (Pandas, Scikit-learn, NumPy), STATA, SQL

Software: Excel (Advanced, VBA, Pivot Tables), Tableau, Power BI, Minitab, SPSS, LaTeX, MS Office Suite (Advance)

Statistics & Methods: Regression (GLM, GAM), Local polynomial regression, Time Series, Survival analysis, Experimental design, Simulation studies, Machine Learning (Scikit-learn, XGBoost, Random Forest, Neural Networks),

Tools & Visualization: ggplot2, plotly, Shiny, highcharter, Matplotlib, Seaborn

Reproducibility: RMarkdown, version control (git), package development (R)

Projects & Research Works

- **Travelers Kaggle Case Competition – Expected Claim Amount Prediction (Finalist, Presentation Round, Dec 2023)**
 - Collaborated in a team to design an auto insurance rating plan for InsNova Auto Insurance using historical claim data.
 - Used ggplot2, shiny, plotly, mgcv, gam in R and matplotlib / seaborn, scikit-learn in python for visualization and modeling.
 - Developed a **two-stage modeling framework**:
 - **Claim Severity: Gamma regression (GLM with offset).**
 - **Claim Frequency: Negative Binomial regression with offset.**
 - Conducted **EDA, variable selection, and feature engineering** (vehicle value category, driver age, vehicle body type, etc.).
 - Evaluated models using **MSE, MAE, and Gini Index**; achieved interpretable and competitive predictive performance.
 - Delivered a final presentation to industry judges, summarizing methodology, assumptions, model performance, and business implications.
 - **GAM analysis of Diabetes Risk Factors – [Git: diabetes project](#)**
 - Applied tensor product smoothing to analyze three-way interactions and interpret nonlinear effects; developed a Shiny App (R) to enable in-time analysis and comparison of different models; generated reproducible reports and plots for presentation.
 - **California Infectious Disease Reporting, 2001 – 2014**
 - Identified predominant infectious disease in California and their causes
 - **Exploring Land Value Disparities in New York City**
 - Built an interactive visual application (shiny app) to compare difference in land prices by borough in New York City.
 - **Public vs Charter Schools in the US**
 - Analyzed the disparities between public and charter schools in US, their causes and effects.
 - Created an interactive dashboard using ShinyApp to display the difference between public and charter schools.
 - **Conformal Prediction on Neural Network, Random Forest and KNN (Tutorial)**
 - Made a tutorial of uncertainty quantification of prediction sets.
 - **A Quasi-Poisson Regression Analysis: Goals in Football, what contributes to them?**
 - Analyzed the predictors that are associated with goal scoring in the top 5 leagues in the world.
 - **Regression Analysis of The Impact of Foreign Direct Investment (FDI) on the Economic Growth of Ghana.**
 - Identified the trend of FDI inflow to Ghana and analyzed its contribution to GDP growth. Results indicated that FDI has had a positive impact in the GDP growth of Ghana.
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Career Experience

University of Massachusetts, Amherst

2022 – Present

Graduate Teaching and Research Assistant | Department of Mathematics and Statistics, UMass Amherst

- Teaching Assistant (Leader) | Stat 240 | Stat 501

- Managed and analyzed performance of 470+ students per semester, identifying trends to improve outcomes.
 - Led weekly discussion for 90+ students per semester; developed clear explanations of complex methods, strengthening communication.
- Research Assistant
- Validated statistical models for accuracy and compliance with specifications.
 - Developed weekly reports to communicate findings for decision-making
 - Conducted time-series and employment trend (2012-2024) producing visuals and summaries.

University of Massachusetts, Amherst

Feb 2023 – May 2024

Statistical Consultant - Lead | Statistical Consultation and Collaboration Services, UMass Amherst

- Compiled datasets analyzed industry-relevant data, and presented actionable recommendations to non-technical stakeholders, a skill directly relevant to actuarial and business analysis.
- Led consulting projects for biomedical and ecological collaborators: prepared datasets, conducted EDA, fit appropriate models (e.g., Beta regression, GLMS), and presented results to non-technical clients.
- Example: Led analysis of molecule permeability into Mycobacterium tuberculosis (Mtb) and smegmatis (Msm); designed comparisons, controlled for covariates, and provided interpretation for wet-lab teams.
- Trained clients in reproducible workflows (R), and documented analysis steps and interpretation for reproducibility and regulatory style reporting.
- Served as the consulting lead in analyzing respiration rate of Crabs.
 - Taught client how to make regression analysis in R and present results.
 - Introduced client to Beta regression and how to interpret its results.

Independent Consultant, Ghana

2020 – 2022

Data Analyst and Consult

Conducted thorough analysis and generated detailed reports on statistical and analytical insights derived from research projects.

- Provided training to individuals on implementing visualization techniques and regression analysis using R.

Kwame Nkrumah University of Science and Technology, Ghana

2019 – 2020

Research and Teaching Assistant | Department of Mathematics, Kumasi

Taught 400+ students how to use statistical tools, such as Excel, R, and SPSS including VBA. Lead tutoring sessions. Administrative activities.

Education

PhD in Statistics (Candidate) | University of Massachusetts, Amherst, May 2024 – Present

- **Qualifying Exams:**

• Oral Qualifying Exam	Passed
• Advanced Statistics v.1	Passed
• Applied Statistics	Passed at the PhD level
• Basic Statistics	Passed at the PhD level
• Basic Probability	Passed at the PhD level
- **Current Research and works:**
 - Improved Bandwidth Selector for Local Polynomial Derivative Estimator (PhD thesis)
 - Extending the dpi bandwidth selector as described by “An Effective Bandwidth Selector for Local Least Squares Regression (Ruppert, Sheather & Wand, 1995)” to a general degree and derivative setting and developing an R package.

Master of Science in Statistics | GPA: 3.95/4.00 | University of Massachusetts, Amherst, Sept 2022 – May 2024

Bachelor of Science in Statistics | CWA: 82.17/100 | Kwame Nkrumah University of Science and Technology, Kumasi, Sep 2015–Jun 2019

Conferences

- New England Student Research Symposium on Statistics and Data Science at Boston University – Student Poster Presentation.
 - 37th New England Statistics Symposium (NESS), May 20-24, 2024, at University of Connecticut – Poster Presentation.
 - NESS-NextGen Data Science Day at Brandeis University, Waltham, MA - Poster Presentation.
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Awards

- First place (graduate and professional category) at the poster competition during the 2025 Data Science Day – NextGen committee of the New England Statistical Society.
 - Outstanding Virtual Poster and Oral Speedy Presentation Presented at the 37th New England Statistics Symposium, May 20-24, 2024.
 - Overall Best Student, Association of Mathematics and Statistics Student – KNUST, Kumasi, 2019
 - Best Statistics Student, Final Year Students, Department of Mathematics and Statistics– KNUST, Kumasi, 2019
 - Best Statistics Student, Third Year Students, Department of Mathematics and Statistics– KNUST, Kumasi, 2018
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Publications

- In Vitro Quality Evaluation of Six Brands of Metformin Hydrochloride Tablets Locally Manufactured in Ghana – *HSOA Journal of Alternative, Complementary & Integrative Medicine* (ISSN: 2470-7562, - 2024)
- Beyond GDP: Unraveling the Nexus of Economic Advancement, Employment Stability, and Forest Conservation in Sub-Saharan Africa - *LNNS Springer Nature, 2023 (Intelligent Computing and Optimization Conference Paper, 2023)*
- Improvement of Grey System Model Using Particle Swarm Optimization. - *Journal of Electrical Systems and Information Technology, Article number: 12 (2021)*